

Team 3: The Wandering Walruses

ES51 Computer-Aided Machine Design

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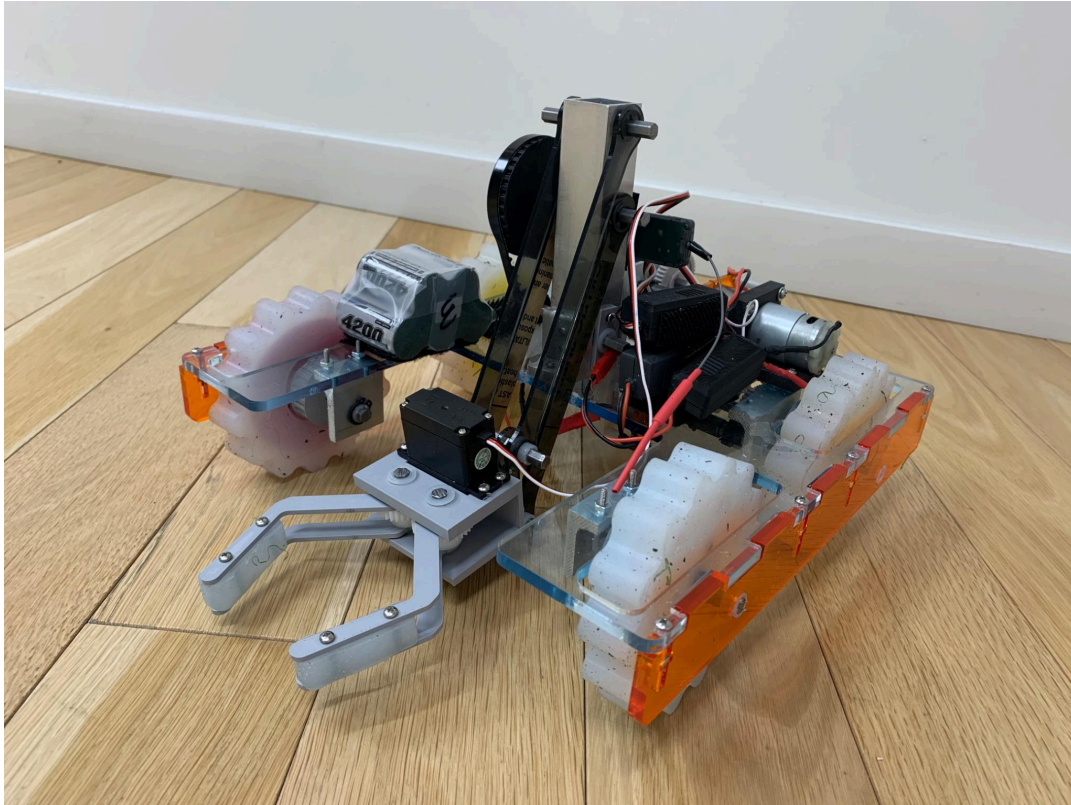


Figure 1: Final Robot

Abstract

The objective of this project was to design, fabricate, and assemble a robot capable of fast, stable movement while collecting and depositing objects like hockey pucks, dog toys, and ping pong balls into scoring zones. Over the course of the semester, the team navigated the ideation phase by sketching multiple ideas, design phase by developing prototypes and a CAD model on Solidworks, manufacturing phase by utilizing various machining techniques to produce the parts, and the testing phase by attempting various challenges and making minor changes to the design. The robot navigated a 15-degree ramp and attempted additional challenges, including a 30-degree incline and an 8.5-inch clearance bridge. It adhered to an 11x11x11-inch size constraint prior to vertical extension. The drivetrain was rear wheel drive, powered by two small, DC motors and gearboxes. A DC motor and pulley mechanism was used to smoothly raise and lower the arm. Finally, a servo motor was used to power the claw, and silicone grips allowed for faster, more reliable scoring. In the final competition, the robot collected one hockey puck and traversed under the bridge, scoring 7 points in both the first and second rounds. In the third round, it scored 11 points by collecting two hockey pucks, two ping pong balls, and driving under the bridge. The robot failed to cross the center, 30-degree ramp, missing critical points.

1. Concept Development

1.1 Problem Statement

The team must design and manufacture a robot capable of picking up objects, securely holding them, and transporting them to baskets upon small ramps to score points. The robot must maintain the maximum height to traverse under the bridge and fit within the “Box of Justice”, and the center of gravity must remain balanced. To achieve these goals, the robot must be easily maneuverable, that can traverse up and down angled ramps, and have a mechanism to firmly grab the objects.

1.2 Constraints

A set of constraints was given at the beginning of the design process that dictated the team’s decisions throughout the semester. The primary constraints were materials, time, and size.

The team was limited to three sheets of ¼-inch acrylic and one sheet of ⅛-inch acrylic. As a result, the prototypes for the base plate, side plates, and arm supports were laser-cut out of cardboard and wood to ensure the correct parameters.

Additionally, the team had a strict timeline to ensure the robot’s completion before impound on December 5, 2024, at 4:00 pm EST. The first several weeks of the term were dedicated to learning engineering and machining techniques, building individual gearboxes, and preparing for the manufacturing process. Approximately three weeks were devoted to manufacturing and assembling additional components for the robot.

Finally, the robot must fit within an 11x11x11-inch cube. As a result, the base plate was made as small as possible. This presented challenges in fitting the motors and gearboxes in their appropriate locations. While strategizing for the competition, the team wanted the robot to be able to score points by driving under the center bridge, so the arm tower was restricted to 8.5 inches tall.

1.3 Ideation

Throughout the ideation phase, the team formulated many solutions to the challenge. The team formed four primary designs for the robot.

Originally, the team sought to focus solely on a robot that could traverse up and down the 30-degree center ramp. Essentially, it would just be a drivetrain with no other mechanisms. This would not test the team’s engineering skills to the fullest, so this design was eliminated. In hindsight, this robot design would’ve gotten to the finals, as the center ramp was worth the most points.

A second idea was a forklift design that would only do the tasks in the four corners and under the center ramp (Figure 2). The team prototyped this idea out of cardboard and realized that this would be difficult to achieve. The primary issue was that the vertical support for the forklift had to be higher than 8.5 inches to drop the pucks and dog toys into their bins, however, then it would not be able to go under the center ramp. The team considered using a mechanism similar to that of a fire truck ladder extending beyond its initial height, but this would be challenging to create with the motors and servo motor. Additionally, the cardboard prototype showed us the uneven center of mass. For these reasons, this design was eliminated.

Another idea was to create tank treads for the wheels and a storage system for the ping pong balls (Figure 3). This idea was eliminated, because treads would be far too difficult to create and implement. Additionally, each ping pong ball was only worth one point, which is not worth the effort it would take to build this design.

The last design was the arm and claw mechanism that was chosen for the final robot. This robot is detailed later in section 3. The arm and claw design was chosen for all the reasons mentioned above, in addition to the feedback received from the design reviews and the team’s Pugh Matrix.

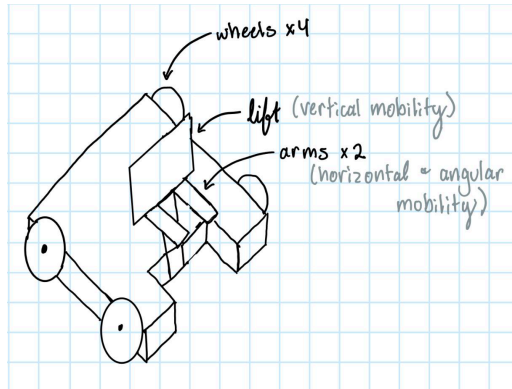


Figure 2: Forklift Design Sketch

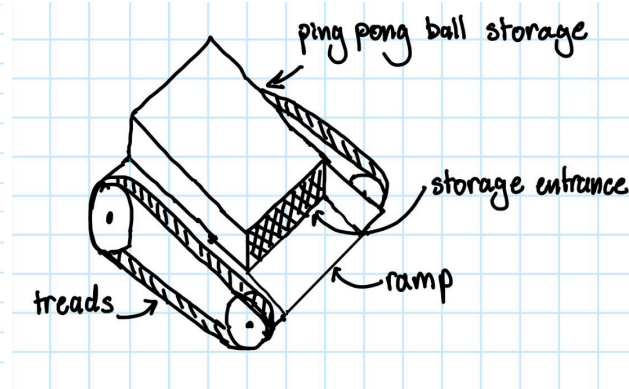


Figure 3: Treads Design Sketch

2. Analysis

2.1 Prototyping

The first means of analyzing the designs was through the creation of prototypes. This allowed the team to look at the general designs in the real world, not on paper, and determine any potential challenges or problems that may arise during the final construction of the robot. It also allowed the team to determine what changes should be made to improve the designs. Additionally, the prototypes ensured that the mechanisms would work as intended when made out of acrylic and aluminum.

The team constructed the first full prototypes out of cardboard. The forklift design (Figure 4) displayed several problems as aforementioned, including the height of the lift and center of gravity. The second full prototype was of the arm and claw mechanism (Figure 5). This prototype showed that a cutout on the front of the baseplate would be needed to ensure the arm could reach the ground to pick up the items. Additionally, prototypes of the arm bars were made of laser-cut wood. With this prototype, the bushings didn't fit exactly into the holes on the bars. As a result, the holes were made larger, allowing the arm to rotate freely. Finally, many prototypes were made of the side and base plates to ensure the correct locations of the holes for the gearboxes, motors, and mechanisms. This process is detailed later in section 2.2.

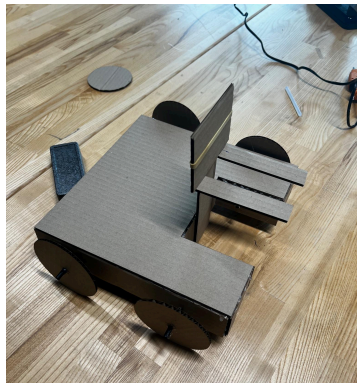


Figure 4: Forklift Prototype

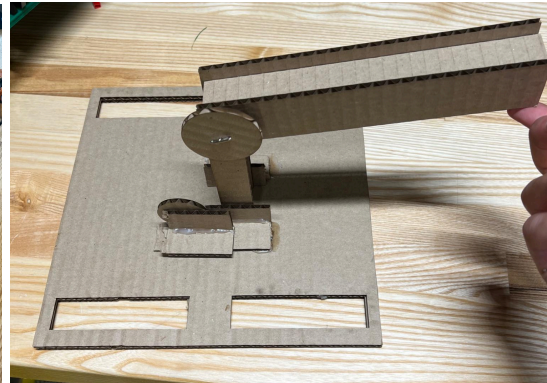


Figure 5: Arm and Claw Prototype

2.2 CAD

A full CAD of the design was created during the design phase. The CAD of the gearbox components was used to determine the location of the holes on the baseplate for all the screws. However, once we laser-cut a cardboard prototype of the base plate, the components didn't fit in the appropriate locations. The team continuously struggled to align the holes of each part on the base plate. A few holes would be shifted to allow one component to fit, however, then the other components wouldn't align. This issue arose since many of the gearbox parts were out of tolerance, so they didn't match the CAD exactly.

The team ended up laser cutting two acrylic base plates, two cardboard base plates, and thirteen cardboard base plates. This same issue arose when cutting the side plates to support the wheels. Approximately twelve iterations of the side plates were used prior to perfecting the design.

However, the CAD was very useful in figuring out the placement of the pulley system to ensure optimal torque to pick up the objects. The team used the CAD to determine the placement of each timing belt pulley, which gave the belt enough tension to provide torque. Additionally, the CAD was used to 3D print parts of the claw and base to hold the servo motor.

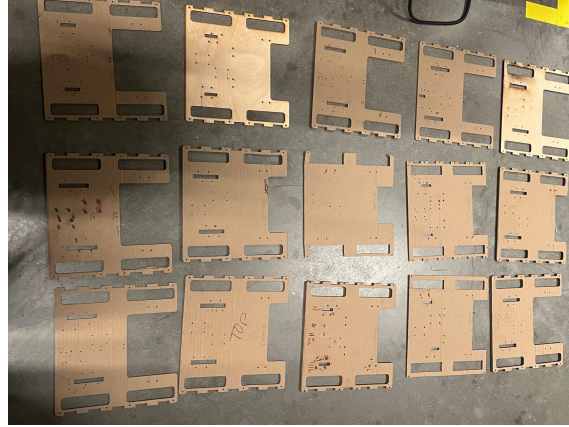


Figure 6: Base Plate Iterations

2.3 Calculations

Throughout the design process, a large concern was the center of mass. First, calculations were performed to ensure that the center of mass was placed in an optimal position to traverse the 30-degree ramp (Figure 7). The center of mass was placed 4.36 inches from the rear wheel point of contact. This position was implemented into the design through the positioning of the gearboxes, arm, and motors. Next, static tests were performed on the ramp. The team moved the arm into various positions to test the maximum height it could be extended before tipping. The claw could be lifted enough to drive up the ramp without interference, but also not too high so the robot would tip backwards.

In the final competition, however, the robot couldn't drive up the center ramp. It wasn't an issue of the center of mass, rather a driving force. The team initially planned for the robot to be pseudo-four wheel drive with a belt connecting the motors on the back wheels to pulleys that would be attached to the front wheel (Figure 8). However, towards the end of the manufacturing process, the belt was too tight to fit onto the front timing belt pulleys. Due to the time constraint, the belts were eliminated from the design to have solely a rear wheel drive robot. A two wheel drive robot was accounted for in the wheel torque calculation, so it should've still been able to drive up the ramp (Figure 9). The calculation was faulty, though, because it didn't account for the slipping of the wheels on the turf that prevented the robot from fully driving up the ramp.

$$\begin{aligned}
 d &= x_{cm} \cos \theta - y_{cm} \sin \theta \\
 &= 5.3 \frac{\sqrt{3}}{2} - \frac{0.45}{2} \\
 &= \boxed{4.36 \text{ in}}
 \end{aligned}$$

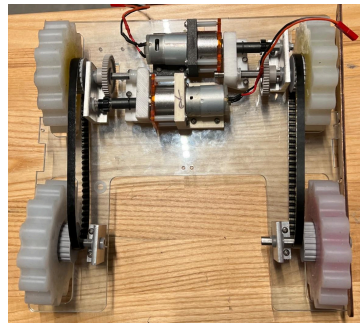


Figure 7:
Center of Mass Calculation

Figure 8:
Drivetrain with Belts

$$\begin{aligned}
 \tau_w &= \frac{mgR}{n} \left(\sin \theta + \frac{v}{gt_a} \right) \\
 &= \frac{mgR}{n} \sin \theta \\
 &= \frac{6.86 * 2}{2} \sin (30^\circ) \\
 &= \boxed{3.43 \text{ lb-in}}
 \end{aligned}$$

Figure 9:
Wheel Torque Calculation

3. Final Solution

3.1 Arm

The arm for the final design was mounted on a tower, made of 1x1 inch square aluminum stock, cut using the horizontal bandsaw (Figure 10). The tower has two holes drill-pressed into it, large enough for the bushings to rotate freely. There are additional slots on the bottom for the tensioning system. The larger slot allows the tensioner to slide, providing different tensions to the arm belt, and the two smaller slots are for screw holes to fasten the tensioner (Figure 11). The tensioner was 3D printed and had a rod through the center with a small 3D printed pulley attached on the end. The middle hole on the tower was made for a rod to attach the large, primary pulley. The belt wrapped around the two pulleys, along with another small pulley on the motor, allowing the arm to be lifted.

For the final robot, however, the tensioner was not needed. The initial design was meant to lock the belt pulley in place, once adjusting the tension of the belt. However, Solidworks was used to determine the position of the tower and the motor to ensure that the exact belt length was fifteen inches, the length of the belt. This made it quite difficult to get the belt on, but the team managed to do it, and the tensioner didn't have to be used. As a result, the belt was tight on the pulleys, without any risk of slippage. One potential concern with the tensioner was that the screws would come loose and tension would be lost. However, the belt was already tight, so even if the tensioner came loose, no tension would be lost.

For the arm itself, four long $\frac{1}{8}$ inch acrylic supports were laser cut. They were connected using two smaller acrylic supports at the end. This ensured that the connected claw would always remain parallel to the ground, allowing for easier pickup of the hockey pucks and dog toys.

The full motor was used, compared to a servo motor, since a high degree of force was needed to lift the arm since the center of mass is far from the base of the robot. An actuator or servo motor wouldn't have been able to effectively lift the arm, especially when the claw was gripping a hockey puck or dog toy.

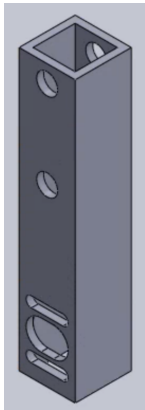


Figure 10:
Arm Tower

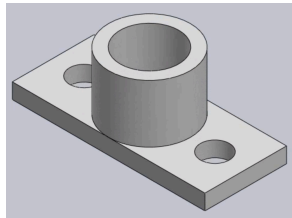


Figure 11:
Tensioner

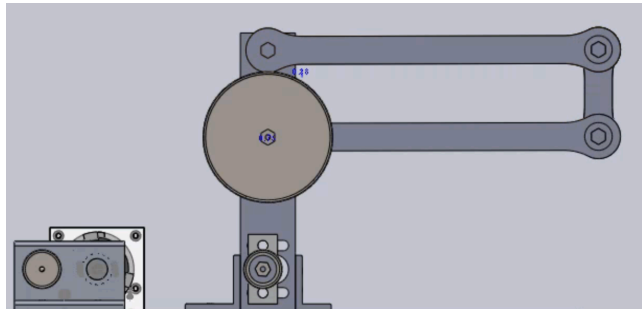


Figure 12:
Final Arm Assembly with Pulleys

3.2 Claw

The claw was attached to the shorter acrylic arm pieces. A frame was designed on Solidworks to connect the claw to the arm, hold the servo motor, and house the gears for the claw. A servo motor was used because only a small amount of force was needed to pick up the hockey pucks and dog toys. All of the parts for the claw were 3D printed, since they were strong enough to hold the objects, acrylic wasn't needed.

However, the PLA claw didn't provide enough force on its own to pick up the objects. To overcome this, silicone inserts were used to increase the coefficient of friction between the surfaces. The A CNC mill was used to create a mold for the inserts. Ecoflex-50, medium-strength silicone was poured into the mold to create the inserts.

All of the parts were strong enough to handle normal operating stresses. However, while testing the robot, it flipped over while going up one of the ramps, since the arm was not lowered. The weight of the robot was placed on the ramp, snapping the acrylic arm pieces approximately 30 minutes before impound. The team had to laser cut new parts, disassemble the arm, and reassemble it completely. Perhaps if 1/4-inch acrylic was used instead of the 1/8-inch acrylic, this problem could've been avoided. If the problem had been avoided, the team may have had enough time to fix the tensioning of the wheels, allowing us to drive up the center ramp and score better in the competition.

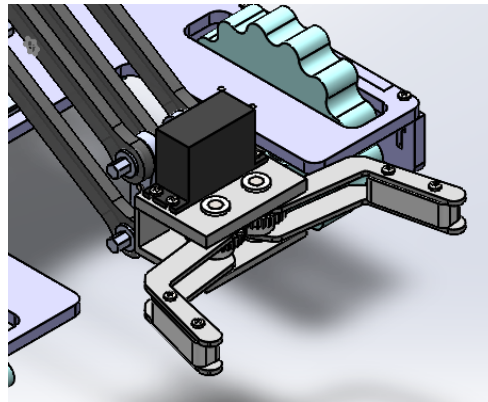


Figure 13: Claw Assembly

3.3 Drivetrain

The robot base and side plates were laser cut out of 1/4-inch acrylic. T-slots were used to connect the side plates to the base plate, in order to support the wheel shafts. The four wheels were cast with Ecoflex 50 silicone, which was molded using a wax mold milled. All of the shafts were turned using the lathe. The gearbox and motor configurations were taken from each team member's individual gearbox project. However, minor changes were made to each of the aluminum and Delrin parts to appropriately use it for the wheels and arm. Additionally, the design was planned to use a belt to connect the powered rear wheels to the front wheels. Under time pressure, however, the team was unable to put the belt onto the wheels and therefore eliminated it from the design.

4. Final Design Specifications

Specification	Value	Justification
Total Height	7.75"	Fitting underneath the bridge
Total Dimensions	10.5"x10.875"x7.75"	Fitting in Box of Justice
Tower Height	4.9375"	Tall enough to go over wall
Total Arm Reach	6.25"	Reducing torque
Arm Reach in Front of Robot	3.75"	Reaching the bin for dropping objects

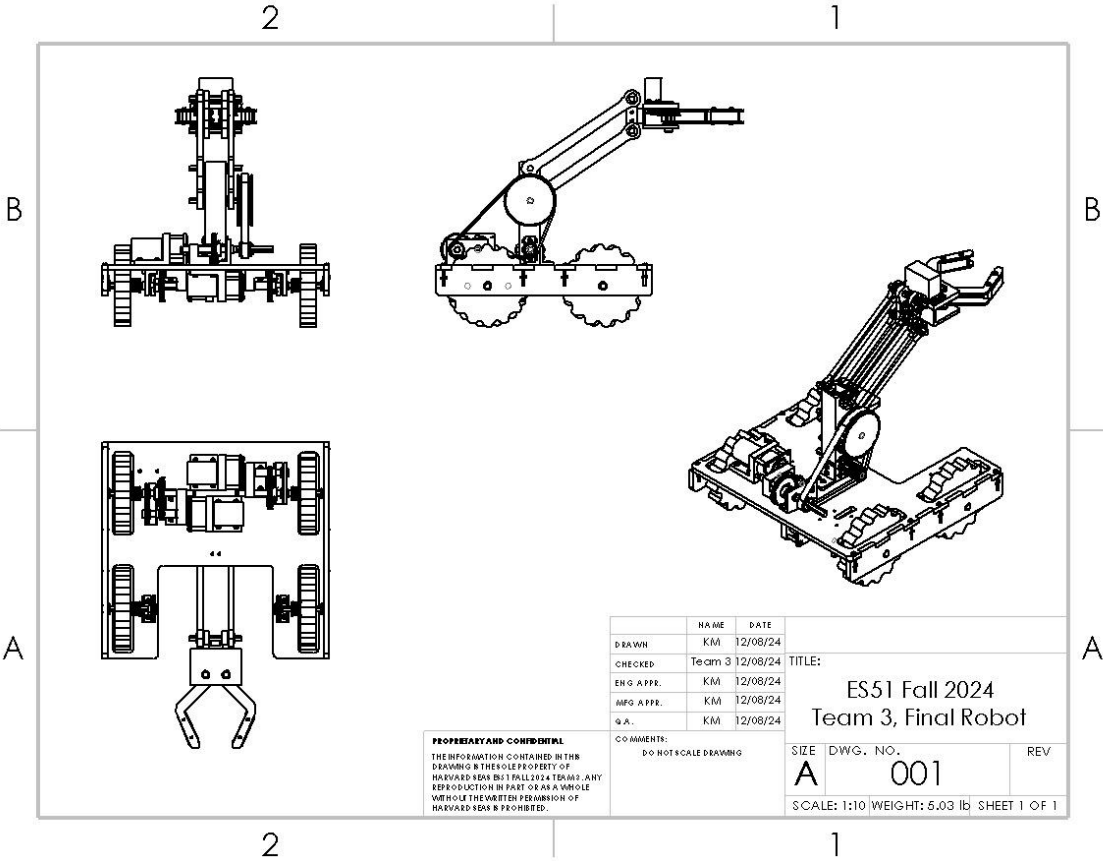
Appendix

Bill of Materials

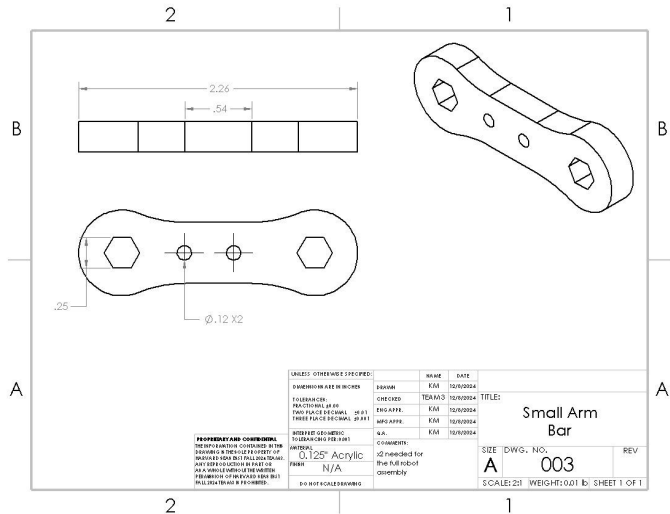
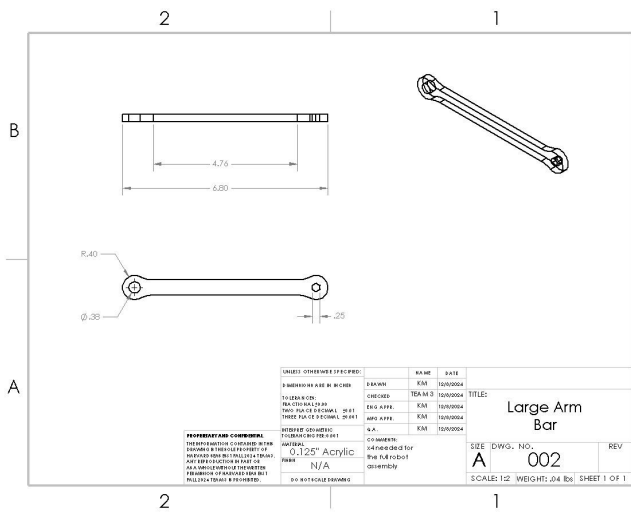
Part Name	Material	Quantity
DRIVETRAIN		
Base Plate	¼" Acrylic	1
Side Plates	¼" Acrylic	2
Wheels	Ecoflex-50 Silicone	4
Front Wheel Hex Shafts	1215 Carbon Steel Hex Rod	2
Rear Wheel Turned Hex Shafts	1215 Carbon Steel Hex Rod	2
Front Wheel Support Bracket	6061 Aluminum Alloy L Bar	2
Support Brackets for Front Axles and Tower	6061 Aluminum Alloy L Bar	4
Nylon Bushing (¼" ID, ⅜" OD)	Nylon	8
Flat Hex Nut (¼" OD)	N/A (for t-slots and front wheel supports)	12
Pan Head 4-40 Screw (7/16")	N/A (for front wheel supports)	8
Pan Head 4-40 Screw (¾")	N/A (for t-slots)	8
Gearbox and Motor Configuration	See Below	2
ARM		
Arm Tower	Aluminum Square Bar	1
Front Tower Support Bracket	6061 Aluminum Alloy L Bar	1
Rear Tower Support Bracket	6061 Aluminum Alloy L Bar	1
Pulley Hex Shafts	1215 Carbon Steel Hex Rod	2
Arm Motor Hex Shaft	1215 Carbon Steel Hex Rod	1
Tensioner	3D Printed PLA	1
Large Timing Belt Pulley (64T)	3D Printed PLA	1
Small Timing Belt Pulley (10T)	3D Printed PLA	2
Belt (15" OD)	Plastic	1
Long Arm Bars	⅛" Acrylic	4
Short Arm Bars	⅛" Acrylic	2
Pan Head 4-40 Screw (⅜")	N/A (for tower supports and tensioner)	12
Flat Hex Nut (¼" OD)	N/A (for tensioner)	4
E Clip (¼" ID)	Metal	4
Gearbox and Motor Configuration	See Below	2
CLAW		
Claw Frame	3D Printed PLA	1
Claw Pieces	3D Printed PLA	4
Claw Gears (32T)	Acetal Plastic	2
Claw Inserts	Ecoflex-50 Silicone	2
Nylon Bushing (¼" ID, ⅜" OD)	Nylon	2
Pan Head 4-40 Screw (¾")	N/A (for silicone claw inserts)	4
Pan Head 4-40 Screw (⅜")	N/A (for servo motor)	4
Servo Motor	N/A	1

GEARBOX AND MOTOR CONFIGURATION		
DC Motor	N/A	1
Motor Mount	3D Printed PLA	1
Unthreaded Spacers (3/16" OD, 11/16" Length)	Plastic	4
Planetary Gearbox	N/A	1
Laser Cut Front Plate	1/8" Acrylic	1
Notched Metal Circle	Metal	1
Axle Support	Delrin	1
Axle Support Shaft	6061 Aluminum Alloy L Bar	1
Hex Drive Shaft	Plastic	1
Small Gear (16T, 32DP)	Acetal Plastic	1
Large Gear (48T, 32DP)	Acetal Plastic	1
Large Gear Round Shaft	6061 Aluminum Alloy Shaft	1
Small Gear Hex Shaft	1215 Carbon Steel Hex Rod	1
Pin (1/16" D, 3/4" Length)	Metal	1
E Clip (1/4" ID)	Metal	1
Nylon Bushing (1/4" ID, 3/8" OD)	Nylon	2
Pan Head 4-40 Screw (7/16")	N/A	8

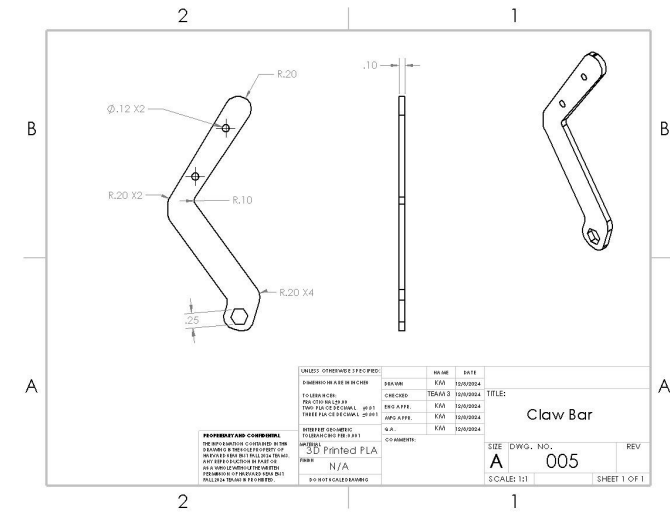
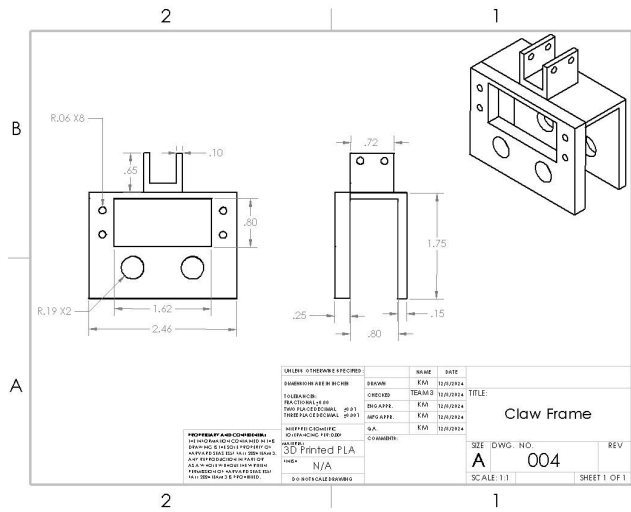
Assembly Drawing



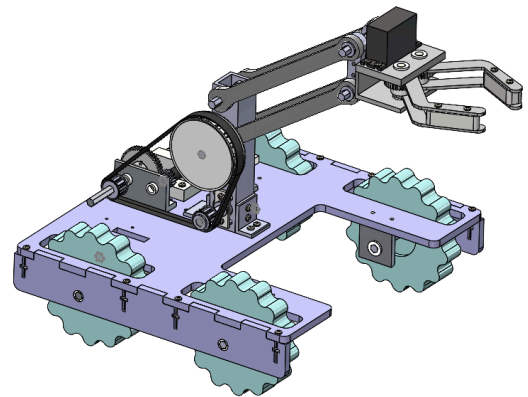
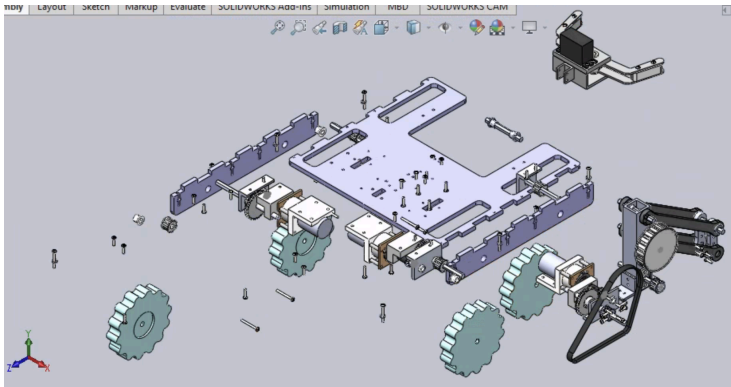
Arm Bars



Claw Pieces



Full CAD



Final Robot

